



(Don't Write anything in this Area)

5.b) Barr body and Karyotype analysis

Barr body refers to the deactivated version of one of 'X' chromosome among 2 'X' chromosome of woman's somatic cells.

While karyotype analysis refers to the analysis of entire sets of ~~one~~ chromosomes in a given cell to identify any deviation in structure & number of chromosomes.

Barr body & Karyotype analysis:

Barr body forms integral part of investigation under karyotype analysis.

as it helps ~~also~~ determine the gender of an individual in definitive terms.

Barr bodies are hence used in Applied anthropology in case of

- Criminal investigation (forensic anthropology)
- Personal identification (" ")
- Sports Anthropology - amongst competitors or athletes to confirm their gender
- Diagnosis of genetic disorders

• Various genetic disorders can be confirmed with kary body study

[ex] Turner's syndrome $\rightarrow 45 X0 \rightarrow$ No barr body in female

Klinefelter's syndrome $\rightarrow 47 XXY \rightarrow$ 1 barr body in male

[Case study] Saunderrajan case - was phenotypically considered as female but and was Indian athlete at global sports competition but later got rejected due to presence of barr body as she was actually diagnosed with ~~human~~ Androgen insensitivity syndrome

Therefore, both techniques have enhanced our understanding of human genetics